Answer Key

Electromagnetic Induction: Lenz's Law & Faraday's Law

> **Given Data:**
> Square loop side $a = 10\,\mathrm{cm} = 0.1\,\mathrm{m}$
> Area $S = a^2 = 0.01\,\mathrm{m}^2$
> Resistance $R = 0.1\,\Omega$
> $\vec{B}$ is perpendicular to the loop plane (Direction: Out of the page).

a) Direction of Induced Current (Lenz's Law)

For $0 < t < 2\,\mathrm{s}$:
The magnitude of the magnetic field B is increasing. Consequently, the magnetic flux Φ through the loop is increasing. According to **Lenz's Law**, the induced current must flow in a direction to create an induced magnetic field $\vec{B}_{ind}$ that opposes the increase in the external flux. Therefore, $\vec{B}_{ind}$ must be opposite to the external field $\vec{B}$.

> *Since $\vec{B}$ points **out of the page**, $\vec{B}_{ind}$ must point **into the page**.*
> *Using the Right Hand Rule, a field pointing into the page corresponds to a **Clockwise (CW)** current.*

For $t > 2\,\mathrm{s}$:
The magnetic field B is constant $(0.2\,\mathrm{T})$. The magnetic flux Φ is constant. There is no variation in flux ($\Delta\Phi = 0$).

$\boxed{\text{Induced Current} = 0}$

b) Expression of B(t) and Magnetic Flux

Choice of Positive Direction: Let's choose the unit normal vector $\vec{n}$ in the same direction as $\vec{B}$ (Out of the page). Thus, $\theta = 0^\circ$ and $\cos(0^\circ) = 1$.
Note: By Right Hand Rule, $\vec{n}$ Out $\implies$ Positive Sense is Counter-Clockwise (CCW).

Expression of $B(t)$:
From the graph description:
For $0 < t < 2\,\mathrm{s}$: $B(t)$ is a linear function passing through the origin ($y = kt$).
Slope $k = \frac{\Delta B}{\Delta t} = \frac{0.2-0}{2-0} = 0.1\,\mathrm{T/s}$.
Thus: $B(t) = 0.1t$ **(SI)**

For $t > 2\,\mathrm{s}$: $B(t) = 0.2\,\mathrm{T}$

Magnetic Flux Φ:
Formula: $\Phi = NBS\cos(\theta)$ where $N = 1$.
$\Phi = 1 \cdot B(t) \cdot 0.01 \cdot 1 = 0.01B(t)$

For $0 < t < 2\,\mathrm{s}$:

$\boxed{\Phi = 0.01(0.1t) = 10^{-3}t\,\mathrm{Wb}}$

For $t > 2\,\mathrm{s}$:

$\boxed{\Phi = 0.01(0.2) = 2 \times 10^{-3}\,\mathrm{Wb}}$

c) Induced e.m.f. and Current

Using Faraday's Law: $e = -\frac{d\Phi}{dt}$

Phase 1 ($0 < t < 2\,\mathrm{s}$):
$e = -\frac{d}{dt}(10^{-3}t) = -10^{-3}\,\mathrm{V}$

$$\boxed{e = -1\,\mathrm{mV}}$$

Induced Current: Using Ohm's Law: $I = \frac{e}{R}$
$I = \frac{-10^{-3}}{0.1} = -10^{-2}\,\mathrm{A}$

$$\boxed{I = -10\,\mathrm{mA}}$$

> ***Comparison:*** *The negative sign indicates the current flows in the negative sense (opposite to our chosen positive sense).*
> *Chosen $\vec{n}$ (Out) $\implies$ Positive Sense is CCW.*
> *Since $I < 0$, the actual current is **Clockwise**.*
> *This confirms the result found in part (a).*

Phase 2 ($t > 2\,\mathrm{s}$):
$e = -\frac{d}{dt}(2 \times 10^{-3}) = 0\,\mathrm{V}$
$\implies I = 0\,\mathrm{A}$ (Consistent with part a).

d) Verification of Direction Independence

Let's choose the positive direction arbitrarily such that $\vec{n}$ is **opposite** to $\vec{B}$.
Now, $\theta = 180^\circ$ and $\cos(180^\circ) = -1$.

New Flux Φ':
$\Phi' = B \cdot S \cdot (-1) = -10^{-3}t$ (for $0 < t < 2$)

New e.m.f e':
$e' = -\frac{d\Phi'}{dt} = -(-10^{-3}) = +10^{-3}\,\mathrm{V}$

New Current I':
$I' = \frac{e'}{R} = +10^{-2}\,\mathrm{A}$

> ***Conclusion:*** *I' is positive, meaning the current flows in the chosen positive direction. However, our new positive direction (associated with $\vec{n}$ opposite to $\vec{B}$) is the reverse of the previous one. Therefore, the physical direction of the current remains unchanged regardless of the arbitrary choice of the normal vector.*